



Basic Non-Sterile Compounding

for Pharmacists and Pharmacy Technicians: Day 2

Speaker: Lcdo. Jorge L. Reyes Quiñones, RPh

Institution: Santa Cruz Compounding Academy

Contact Hours: 6.0 | **Application Level:** 1 (Introductory)

Colegio de Farmacéuticos de Puerto Rico · Continuing Education Division
PO Box 360206, San Juan PR 00936-0206 · Tel. (787) 754-8649 · cfpr.org

ACPE Provider Number: 0151

Disclosure to Participants

ACPE Standards 2015 — Conflict of Interest Disclosure

Lcdo. Jorge L. Reyes Quiñones, RPh, faculty for this CE activity, has no relevant financial relationships to disclose with ineligible companies.

- The content of this activity was developed without commercial influence.
- The presentation contains no commercial logos or promotional material.
- Complies with ACPE Standards 2015 — conflict of interest disclosure.
- Any relevant financial relationship has been mitigated prior to this activity.

ACPE Accreditation Statement

Continuing Pharmacy Education Provider

"The Colegio de Farmacéuticos de Puerto Rico is accredited by the Accreditation Council for Pharmacy Education as a provider of continuing pharmacy education."



Provider Number	0151
Contact Hours (Day 2)	6.0 · Full course: 18 hours / 1.8 CEUs
Knowledge Level	1 (Introductory) — Application-Based
Deadline to claim credits	45 days from the event
Validation	cfpr.org + CPE Monitor

General Rules and Regulations

ACPE-Accredited Continuing Education

1. Every participant must have registered their attendance.
2. Participants must remain physically present for the entire activity or until the speakers finish their presentation, since **no partial credits are awarded**.
3. The grace period for (2.0)-hour lectures is **(20) minutes**. On-site registration is not accepted after the corresponding grace period has ended.
4. Guests are not allowed in the lecture rooms.
5. The use of cell phones is not allowed in the lecture rooms. Please set them to silent mode; if you need to answer a call, please step out of the room to avoid disturbing your colleagues.
6. Recording the lecture is not allowed, neither video nor audio.
7. Remember that you have **(45) days** from the date of each lecture to claim your credits.
8. To claim your credits you must complete the Evaluation and the Test, which must be passed with **more than 70%**.
9. Log in at **www.cfpr.org**, find this lecture in the Course History and click the icon. First the Evaluation; once completed, the system authorizes taking the Test. After both steps are completed, the Certificate is released.
10. To claim credits with **ACPE / NABP**, access the **CPE Monitor** (through www.cfpr.org) and enter the **Access Code** provided at the end of the activity.

Objectives — Day 2

Upon completion of the CPE, the participant will be able to:

- 1 Review the calculations, formulation, procedures, and equipment used to prepare suppositories.
- 2 Differentiate rectal, vaginal, and urethral suppositories.
- 3 Compound rectal and vaginal suppositories.
- 4 Review the indications, calculations, procedures, and devices used to prepare dermatological formulations.
- 5 Prepare two dermatological formulations and package them in different devices.

Review — Day 1

What we covered yesterday in 6 key points

1

Compounding definition: 503A vs. 503B (DQSA 2013).

2

USP <795> Nov 2023: key BUDs (14/35/90/180 days).

3

USP <800>: Hazardous Drugs, PPE, engineering controls.

4

PR Act No. 247-2004 + Regulation 156 (Board of Pharmacy).

5

Calculations: potency, displacement value, fill weight.

6

Lab Day 1: 100 SR capsules, weight deviation $\pm 7.5\%$.

Source: DQSA 2013 · USP <795> 2023 · USP <800> · Act 247-2004

Instructional Methodology

Time distribution — Day 2 (6 contact hours)

- **Audiovisual lecture:** suppositories (calculations, anatomy, bases), dermatologicals.
- **Active learning:** 2 pop questions + 2 clinical cases (1 suppository, 1 dermatological).
- **Hands-on laboratory:** 12 rectal suppositories + 2 dermatological formulations in different devices.

Block	Duration
Introduction + Disclosure + Day 1 Review	60 min
Module 1: Suppositories (calculations, bases, anatomy)	90 min
Module 2: Dermatological formulations + devices	90 min
Laboratory: suppositories + 2 dermatologicals	60 min
Post-test + Activity Evaluation Form	60 min

Needs Assessment

Needs analysis — why this course?

Why does it matter?	What to expect after the course?	What the participant should know	Application Activity
~3–5% of all compounding is suppositories (high demand)	Safe preparation of rectal/vaginal suppositories	DF, bases (cocoa butter, Witepsol, PEG), +10% overage	12 Diclofenac rectal suppositories + weight control $\pm 5\%$
25–30% of US compounding is dermatological (PCCA 2023)	Dermatological formulation with the correct vehicle	Creams vs ointments vs gels, indications, devices	Depigmenting cream + salicylic acid ointment in different containers
Melasma, alopecia, neuropathic pain — personalized Rx	Combination mixtures (KAL, Kligman) compounded	Geometric dilution, levigation, EMP/Unguator, ointment mill	Combination cream with label + documented BUD

Source: NABP 2023 · ACPE Needs Assessment Guidelines · USP <795>/<800> · PR Act 247-2004

Activity Planning

Objectives · Practices · Final product as evidence

Objective	Practice / Experience 1	Practice / Experience 2	Final Product (Application Activity)
1. Review suppository calculations + procedures	Solve DF + excipient formula	Example Diclofenac 50 mg 12 supp + scale	Complete calculation sheet
2. Define/differentiate rectal/vaginal/urethral	Compare anatomy + absorption	Identify base by suppository type	Documented comparison table
3. Compound rectal/vaginal suppositories	Fusion + molding step by step	Weight control $\pm 5\%$ + visual inspection	12 packaged Diclofenac rectal suppositories
4. Review dermatological calculations + indications	% per batch calculation + CoA adjustment	Levigation + geometric dilution	Calculation sheet + vehicle selection
5. Prepare 2 dermatological formulations	Depigmenting cream (Hydroquinone/Salicylic Acid/Ascorbic Acid/Betamethasone) 45 g	Salicylic Acid 10% in White Petrolatum (W.P.) 100 g	2 products in different devices

Source: ACPE Standards 2015 · Bloom's Taxonomy (Application Level) · USP <795>/<800>

Competencies Assessed

Apply equally to pharmacists, technicians and students

Competencies

1. Knowledge of fundamentals
2. Patient-centered care
3. Use of Medications
4. Problem solving
5. Innovation
6. Professionalism

Post-test $\geq 70\%$ required — uniform assessment for all profiles

Suppositories — Calculations (1/2)

Displacement Factor (DF) and excipient formula

Displacement Factor (DF)

DF = weight of excipient displaced by 1 g of API.

Formula: Excipient = (Total weight × n suppositories) – (API quantity ÷ DF) × n

Example — Indomethacin 50 mg rectal suppository

Rx: 12 indomethacin 50 mg suppositories in cocoa butter base.

Cocoa butter DF for indomethacin = 1.1 · Fill weight (mold #1) = 2.0 g.

Excipient = (2.0 × 12) – (0.050 ÷ 1.1) × 12 = 24 – 0.545 = 23.455 g cocoa butter.

Double casting: technique for APIs whose DF differs greatly from the excipient (2 molds, double fusion).

Suppositories — Procedures (2/2)

Weight control, storage, BUD

Weight control (in-process):

- Weigh each suppository: limit $\pm 5\%$ of the theoretical weight.
- Visual inspection: color uniformity, smooth surface, no bubbles/cracks.

Storage:

- Refrigeration 2–8°C: cocoa butter, low-MW PEG.
- Room temperature: firm PEGs and stable Witepsol.

Default USP <795> 2023 BUD: Nonaqueous suppositories (cocoa butter, Witepsol, anhydrous PEG) without stability data:

180 days (refrigerated or room temperature depending on the base).

Source: USP <795> 2023 · PCCA · Allen's IJPC

Equipment and Molds

Suppository molds + laboratory support

Mold type	Typical weight	Comment
Standard rectal (adult)	1 – 3 g	Torpedo / ogival
Pediatric rectal	0.5 – 1 g	Reduced size
Vaginal (pessary)	2 – 5 g	Globular / oval
Urethral (bougie)	1 – 4 g	Elongated cylindrical

Materials: aluminum/metal (multi-use, wash and dry), disposable plastic, silicone.

Pre-lubrication: light USP mineral oil, per compatibility with the base.

Laboratory support: analytical balance (Class I/II, ± 0.1 mg, NIST calibration), hotplate with calibrated thermometer, 2–8°C refrigerator with a daily log, spatulas, glass beakers, waxed paper.

Source: USP <795> 2023 · PCCA Equipment Guide · Allen's

Anatomy and Absorption

Rectal, vaginal and urethral

Rectal	Vaginal	Urethral
• Superior hemorrhoidal veins → portal vein (hepatic metabolism). • Middle / inferior veins → vena cava (partial hepatic bypass). • pH: 7.2–7.4. No active digestive enzymes. • Indications: analgesia, antiemetic, anticonvulsant.	• Absorption through the vascularized vaginal epithelium. • Normal pH: 3.5–4.5 (protective acidity). • Vaginal T° ~37°C — critical melting point. • Indications: antifungals, antibacterials, estrogens, progesterone.	• Elongated cylindrical shape (bougie). • Limited use in daily practice. • Examples: alprostadil (MUSE), sildenafil. • Bases: high-MW PEG (firm consistency).

Source: Ansel's 2014 · Remington 22nd Ed. · USP <795> 2023 · PCCA

Types of Suppositories

Comparison table of the 3 dosage forms

Feature	Rectal	Vaginal	Urethral
Shape	Torpedo / ogival	Pessary / globular	Elongated cylindrical
Typical weight	1–3 g adult · 0.5–1 g ped.	2–5 g	1–4 g
Temperature	≤ 37°C	≤ 37°C	Firm at room T°
Preferred base	Cocoa butter, Witepsol	PEG, polymers	High-MW PEG
Insertion	Deep rectum	Vaginal + applicator	Urethra
Absorption	Systemic + local	Predominantly local	Local

Source: USP <795> 2023 · Ansel's · PCCA · Allen's IJPC

QUICK QUESTION #1 · Learning Check

Suppository Bases — Melting Point

Which suppository base has a melting point of ~34–36°C and may NOT be ideal in a tropical or febrile setting (vaginal use)?

A) PEG 3350

B) Witepsol H15

C) Cocoa butter (theobroma oil)

D) Gelatinized glycerin

ANSWER — QUICK QUESTION #1

Suppository Bases — Melting Point

Which suppository base has a melting point of ~34–36°C and may NOT be ideal in a tropical or febrile setting (vaginal use)?

A) PEG 3350

B) Witepsol H15

C) Cocoa butter (theobroma oil) ✓ **CORRECT**

D) Gelatinized glycerin

Rationale: Cocoa butter melts at 34–36°C. In the vagina (~37°C) or with fever it may soften before dissolving. PEG and Witepsol are more stable.

Suppository Bases (1/2)

Cocoa Butter and PEG

Cocoa Butter (Theobroma Oil)

- Melting point: 34–36°C.
- Rapid melting at body T°, biocompatible.
- Polymorphism (α , β , γ forms — β' desired).
- Unstable if overheated.
- Refrigerated storage (2–8°C).
- BUD: 180 days (USP <795> 2023).

PEG (Polyethylene glycol)

- Grades: PEG 400 (liquid) – PEG 6000 (solid).
- Mechanism: dissolution (not melting).
- Stable, requires no refrigeration.
- Compatible with a broad range of APIs.
- Hygroscopic — may irritate mucous membranes.
- Typical formula: PEG 1000 75% + PEG 3350 25%.

Source: USP <795> 2023 · PCCA Suppository Bases · Allen's · Remington

Suppository Bases (2/2)

Witepsol and Gelatinized Glycerin

Witepsol (Adeps Solidus)

- Lauric acid esters (H, W, E).
- Melting point: 33.5–44°C depending on grade.
- High stability, non-polymorphic.
- Compatible with more APIs than cocoa butter.
- Witepsol H15: the most common (~34–36°C).
- Preferred for industrial use.

Gelatinized Glycerin

- Water-soluble base: gelatin + glycerin + water.
- Vaginal and urethral use.
- Does not melt — dissolves in body fluids.
- Less common in modern compounding.
- Refrigerated to preserve consistency.

Source: USP <795> 2023 · PCCA · Allen's · Ansel's 2014

Procedure — Compounding Suppositories

12 rectal suppositories step by step (Obj 3)

- 1 Verify MFR, Rx, CoA, fill weight, DF, BUD.
- 2 PPE + calibrated balance + lubricated molds (mineral oil).
- 3 Weigh API and excipient (double-checked with a 2nd operator).
- 4 Melt the base at minimum T° (cocoa butter 35–37°C). Do NOT overheat.
- 5 Cool to ~32°C before incorporating a heat-labile API.
- 6 Pour into molds slightly above the rim (contraction).
- 7 Refrigerate 30–60 min (2–8°C) until fully solidified.
- 8 Unmold, weigh ($\pm 5\%$), inspect, package, label per Act 247.

Source: USP <795> 2023 · PCCA SOP Suppositories · Allen's · Remington

Clinical Case #1 — Rectal Suppository

Clinical application with patient description

Patient description

52-year-old woman with severe rheumatoid arthritis. Post-operative from abdominal surgery, with paralytic ileus — oral route not tolerated. The rheumatologist prescribes: Indomethacin rectal suppository 50 mg q12h × 5 days. No commercial rectal indomethacin presentation exists in PR → 503A compounding indicated.

Compounding plan

DF: indomethacin in cocoa butter = 1.1.

Fill weight: adult rectal mold = 2.0 g.

Batch: 12 suppositories (5 days × 2/day + 2 extra).

Calculation: Cocoa butter = $(2.0 \times 12) - (0.050 \div 1.1) \times 12 = 23.455$ g.

Counseling: insertion, storage 2–8°C, proper disposal, local effects.

Source: USP <795> 2023 · Allen's IJPC · PCCA · Ansel's 2014

Dermatological Formulations (1/2)

Calculations and incorporation techniques (Obj 4)

Percentage formula → quantity per batch

$\% \times \text{total batch weight} = \text{component quantity}$. E.g.: Hydrocortisone 2% in 60 g → $0.02 \times 60 = 1.2$ g.

Potency adjustment (CoA)

$\text{API quantity} = (\text{desired \%} \times \text{batch}) \div (\text{potency} \div 100)$.

Incorporation techniques

- **Levigation:** reduce particle size with a levigating agent (mineral oil, glycerin) before incorporating.
- **Geometric dilution:** progressive dilution in a mortar for potent APIs.
- **Fusion:** melt bases on a hotplate, cool, incorporate the API, homogenize.
- **Mechanical mixing:** Unguator, DAB Mixer, Luer-lock syringes for gels.

Source: USP <795> 2023 · PCCA Dermatology 2023 · Allen's · Remington

Dermatological Formulations (2/2)

Clinical indications and devices

Main indications

- Eczema / atopic dermatitis: corticosteroids, tacrolimus, emollients.
- Psoriasis: coal tar, salicylic acid, calcipotriol, anthralin.
- Topical infections: metronidazole, mupirocin, clotrimazole.
- Neuropathic pain: KAL cream (ketamine + amitriptyline + lidocaine).
- Scars: silicone gel, onion extract, vitamin C.
- Alopecia: minoxidil 5% foam, topical finasteride.

Packaging devices

- Jars: creams and dense ointments — easy access.
- Airless pump: fluid creams / antioxidants — hygienic.
- Calibrated syringe: dense gels, dosing in grams.
- Vaginal/rectal applicators: targeted application.
- Squeeze tubes: continuous creams/gels.
- Foil pouches: unit doses — maximum stability.

Source: USP <795> 2023 · USP <661> · PCCA Dermatology 2023 · Allen's

Dermatological Clinical Indications

Table of conditions + compounded formulation

Condition	API Compounded	Vehicle	Device
Severe eczema	Tacrolimus 0.1%	Base cream	Jar / airless
Psoriasis	Salicylic acid 6% + betamethasone	Ointment	Jar / tube
Melasma (hyperpigmentation)	Kligman: HQ 4% + tretinoin 0.025% + hydrocortisone 1%	Base cream	Opaque jar
Androgenetic alopecia	Minoxidil 5% + finasteride 0.1%	Foam / gel	Airless pump
Neuropathic pain	KAL Cream (K10/A4/L2)	Lipoderm / PLO	Syringe / airless
Vitiligo	Tacrolimus 0.1% + vit. D3	Base cream	Jar
Severe acne	Isotretinoin 0.05% + clindamycin 1%	Gel	Airless pump
Scars	Silicone 10% + vit. E	Gel	Squeeze tube

Source: PCCA Dermatology 2023 · AAD Guidelines · Allen's · Remington

Dermatological Vehicles

Selection by clinical condition

Ointment	Oily base (petrolatum, lanolin). Greater occlusivity → greater absorption. For very dry skin, ulcers.
Cream	O/W or W/O emulsion. Easy application, better accepted. For dermatitis, infections, face, hairy areas.
Gel	Hydrophilic base (carbopol, HPMC). Cooling, fast drying. For acne, alopecia, topical pain.
Foam	Minimal density, easy application. For the scalp (minoxidil), hairy areas.
PLO Gel	Pluronic Lecithin Organogel. Transdermal penetration. For neuropathic pain, TD anti-inflammatories.
Lipoderm	PCCA transdermal vehicle. Excellent penetration, well tolerated. For hormones, TD ketamine.

Source: Ansel's 2014 · USP <795> 2023 · PCCA Vehicle Guide · Allen's

Packaging Devices — Compatibility

USP <661> Containers — Plastics

Device	Compatible formulations	Main advantage
Jar HDPE	Ointments, dense creams	Easy access, low cost
Airless pump	Fluid creams, gels, antioxidants	Hygienic, minimal oxidation
Calibrated syringe	Dense gels, targeted creams	Precise dosing (g)
Vaginal applicator	Vaginal creams / gels	Targeted insertion
Squeeze tube	Continuous creams / gels	Portable, no finger contact
Foil pouch	Unit doses	Maximum stability

Mandatory rules: USP <661> requires container-API compatibility (pH, T°, composition). Act 247 + Regulation 156 require a complete label: Rx, BUD, storage, concentration, patient.

Source: USP <661> · USP <795> 2023 · PCCA Packaging · PR Act 247-2004

QUICK QUESTION #2 · Learning Check

Dermatological Vehicle — Selection

For a minoxidil 5% formulation intended for the scalp in a patient with androgenetic alopecia, which vehicle is MOST appropriate?

A) 100% petrolatum ointment

B) Hydroalcoholic foam or gel

C) Dense W/O cream with lanolin

D) Opaque paste with starch

Dermatological Case — Melasma

Facial hyperpigmentation · Modified Kligman formula

Patient description

38-year-old woman with post-pregnancy facial melasma (chloasma) refractory to sunscreen and OTC depigmenting agents. The dermatologist prescribes a modified Kligman formula in cream; 30 g. Apply a thin layer to hyperpigmented areas once daily (at night). Strict daytime photoprotection.

Component	% (30 g)	Function / Notes
Hydroquinone (HQ)	4% → 1.2 g	Depigmenting — inhibits tyrosinase
Tretinoin	0.025% → 7.5 mg	Renews the epidermis, increases HQ penetration
Hydrocortisone	1% → 300 mg	Anti-inflammatory — reduces HQ/tretinoin dermatitis
Hydrophilic base cream	q.s. → 28.5 g	Vehicle

Critical counseling: Apply ONLY at night · avoid mucous membranes and eyelids · SPF 50+ photoprotection mandatory · maximum use 4–6 months · pregnancy = CI (HQ + tretinoin). Packaging: opaque jar with an airtight lid (HQ oxidizes in light).

ANSWER — QUICK QUESTION #2

Dermatological Vehicle — Selection

For a minoxidil 5% formulation intended for the scalp in a patient with androgenetic alopecia, which vehicle is MOST appropriate?

- A) 100% petrolatum ointment
- B) Hydroalcoholic foam or gel ✓ CORRECT**
- C) Dense W/O cream with lanolin
- D) Opaque paste with starch

Rationale: the scalp requires a non-greasy vehicle that is easy to apply and dries quickly. Foams and hydroalcoholic gels are the clinically preferred option.

Clinical Case #2 — Dermatological

Clinical application with patient description

Patient description

48-year-old man with refractory painful diabetic neuropathy. Oral gabapentin causes severe drowsiness. The neurologist prescribes: KAL Cream (Ketamine 10% / Amitriptyline 4% / Lidocaine 2%) in Lipoderm® 30 g, apply BID to the feet. 503A compounding indicated.

Compounding plan

Quantities: Ketamine 3.0 g · Amitriptyline 1.2 g · Lidocaine 0.6 g · Lipoderm q.s. 30 g.

Technique: levigation + geometric incorporation + Unguator for homogeneity.

Packaging: airless pump 30 mL. BUD 90 days nonaqueous (USP <795> 2023).

Counseling: gloves when applying, avoid mucous membranes, wash hands, do not apply on broken skin.

Source: PCCA KAL · USP <795> 2023 · IJPC · Allen's

Summary — Day 2

Six key points of the module

1

Suppositories: calculations (DF) + procedures + bases (cocoa, PEG, Witepsol).

2

Differentiating rectal / vaginal / urethral (anatomy, weight, base, absorption).

3

Compound rectal/vaginal — weight control $\pm 5\%$, BUD 180 days.

4

Dermatologicals: calculations, indications, vehicles and devices.

5

Lab: 2 dermatological formulations + packaging in different devices.

6

CPE reminder: post-test $\geq 70\%$, 45 days, [cfpr.org](https://www.cfpr.org) + CPE Monitor.

Suppositories — Excipients and Overage

Silicon dioxide + the +10% calculation rule

Silicon Dioxide (SiO₂)

Function: suspending agent.

Purpose in suppositories: helps distribute the API uniformly throughout the entire suppository mass, preventing sedimentation during molding and cooling.

It does NOT increase the solubility of the API.

It does NOT adjust the pH or affect metabolism.

- Typical concentration: 0.5–2%.

Overage Rule (+10%)

- **ALWAYS** calculate an extra 10% to compensate for losses during molding, scraping and QC.
- E.g.: Rx calls for 30 suppositories → calculate for 33.
- E.g.: Rx calls for 12 suppositories → calculate for 13 (minimum 1 extra).
- The overage prevents producing incomplete suppositories due to contraction on cooling.

Progesterone Suppositories — Pregnancy

Validated clinical indications (vaginal / rectal)

Micronized progesterone compounded into vaginal suppositories is prescribed in obstetrics for multiple indications:

Obstetric indication	Clinical description
Prevention of spontaneous abortion (high-risk)	Luteal insufficiency, recurrent miscarriage — support in the 1st trimester
Reduction of preterm birth	History of preterm delivery, short cervix < 25 mm — 200 mg vaginally QHS
Support after fertility treatment (IVF)	Luteal phase after IVF — support for embryo implantation
Luteal phase support	Primary luteal insufficiency — cyclic support
ALL are recognized indications. Typical dose: 100–400 mg micronized progesterone vaginally QHS · Witepsol or PEG base · BUD 180 days refrigerated.	

Source: ACOG Practice Bulletin · NIH/NICHD · Allen's · PCCA Women's Health

Advanced Equipment — Mill / EMP

Particle reduction + homogeneous mixing

Equipment	Main function	Typical use
Ointment Mill (3-roller)	Reduce particle size	Creams, dense ointments, TD hormonals
EMP (Electronic Mortar and Pestle / Unguator®)	Mechanical mixing and homogenization	Suspensions, creams, ointments — in their final container
V-Blender Pro	Dry powder mixing	Capsules, pre-blend before filling
Hotplate stirrer	Fusion + stirring	Suppository bases, gels
EMP — Electronic Mortar and Pestle • Common commercial brand: Unguator® (GAKO). • Mixes in the final container (jar, airless) → no loss and no cross-contamination. • Programmable speeds; reduces operator variability. • Preferred in BHRT, compounding dermatology and pediatric suspensions.		

Source: USP <795> 2023 · PCCA Equipment · Allen's · Unguator manufacturer

Low-Dose Naltrexone (LDN)

Immunomodulatory compounding of growing clinical use

Naltrexone is an FDA-approved opioid antagonist at 50 mg (alcohol/opioid). **At a low dose (LDN: 1.5–4.5 mg/day)** it is postulated to modulate the immune system and reduce inflammation.

Reported indications (compounded LDN)

- Hailey-Hailey disease (benign familial pemphigus).
- Refractory psoriasis.
- Scleroderma.
- Fibromyalgia, chronic fatigue syndrome.
- Crohn's disease / ulcerative colitis (off-label use).

Common form: compounded capsules of 1.5/3.0/4.5 mg (not commercially available at these doses).

Source: LDN Research Trust · PCCA · IJPC · Allen's · BMC Dermatology 2018

Alopecia — Advanced Compounding

Minoxidil + finasteride + dutasteride — combination options

Androgenetic hair loss can be treated with compounding by combining APIs with different mechanisms:

API	Mechanism	Typical topical dose
Minoxidil	Vasodilator / prolongs the anagen phase	2–5% solution, foam or gel
Finasteride	Type II 5α-reductase inhibitor	0.05–0.25% in gel or foam
Dutasteride	Type I and II 5α-reductase inhibitor (more potent)	0.01–0.1% in gel or solution
Tretinoin (adjuvant)	Renews the follicle + increases minoxidil penetration	0.01–0.025% in combination
Caffeine	Follicular stimulant (adjuvant use)	0.1–0.5%
Common combo: Minoxidil 5% + Finasteride 0.1% + Dutasteride 0.05% in hydroalcoholic foam — preferred in patients refractory to monotherapy.		

Source: AAD Hair Loss Guidelines · PCCA Hair Loss Compounding · Allen’s · IJPC

POST-TEST

Day 2 · 15 questions
Select the BEST answer
Passing score required: $\geq 70\%$
One (1) question per slide

POST-TEST · Question 1 / 15

Suppositories — BUD

Most suppositories have an assigned BUD of:

A) 30 days

B) 60 days

C) 90 days

D) 180 days

E) none of the above is correct

Source: USP <795> 2023

POST-TEST · Question 2 / 15

Silicon Dioxide in Suppositories

The use of silicon dioxide in compounding suppositories is that, as a suspending agent, it:

- A)** helps distribute the API throughout the entire suppository mass
- B)** increases the solubility of the API
- C)** adjusts the pH of the preparation
- D)** improves the metabolism of the API
- E)** none of the above is correct

Source: Allen's · PCCA Suppository Guide

POST-TEST · Question 3 / 15

Suppository Overage

If you have a prescription for 30 suppositories, it is recommended to calculate for:

- A)** 30 suppositories
- B)** 31 suppositories
- C)** 33 suppositories
- D)** 40 suppositories
- E)** none of the above is correct

Source: PCCA · Allen's — +10% overage rule

POST-TEST · Question 4 / 15

Storage of Suppositories

Suppositories are best stored:

- A) at room temperature
- B) in the refrigerator
- C) in the freezer
- D) between 45° and 50°C
- E) none of the above is correct

Source: USP <795> 2023 · Allen's

POST-TEST · Question 5 / 15

Progesterone in Pregnancy

Progesterone suppositories are used during pregnancy to:

- A)** prevent spontaneous abortion in high-risk pregnancies
- B)** reduce the risk of preterm birth
- C)** provide support after fertility treatments
- D)** provide luteal phase support
- E)** all of the above are correct

Source: ACOG · NIH/NICHD · PCCA Women's Health

POST-TEST · Question 6 / 15

Dermatological Conditions

Dermatological condition(s) commonly treated with a compounded preparation:

A) acne

B) psoriasis

C) melasma

D) hair loss

E) all of the above are correct

Source: AAD · PCCA Dermatology

POST-TEST · Question 7 / 15

Mill Equipment

The mill is equipment used to:

- A)** reduce the particle size of the ingredients
- B)** mix all the ingredients very homogeneously
- C)** distribute the API in the diluent
- D)** all of the above are correct
- E)** none of the above is correct

Source: USP <795> 2023 · PCCA Equipment Guide

POST-TEST · Question 8 / 15

EMP Acronym

The acronym EMP stands for:

- A) equilibrate the mass and the pressure
- B) electronic mortar and pestle
- C) electronic manufacturing process
- D) the combination of estradiol and micronized progesterone
- E) none of the above is correct

Source: PCCA Women's Health Compounding Guide · IJPC

POST-TEST · Question 9 / 15

Low-Dose Naltrexone

Low-dose naltrexone can help patients with:

- A)** Hailey-Hailey disease
- B)** psoriasis
- C)** scleroderma
- D)** all of the above are correct
- E)** none of the above is correct

Source: LDN Research Trust · IJPC · BMC Dermatology 2018

POST-TEST · Question 10 / 15

Hair Loss Treatment

Hair loss can be treated with:

- A) minoxidil
- B) finasteride
- C) dutasteride
- D) all of the above are correct
- E) none of the above is correct

Source: AAD Hair Loss Guidelines · PCCA

POST-TEST · Question 11 / 15

Hydroquinone Rx — Indication

Rx: Hydroquinone 8% · Salicylic Acid 2% · Vit. C 550 mg × 2 · Betamethasone Valerate 15 G · Emollient Cream 45 G · Mix and prepare this cream · Sig: apply as directed. Use this prescription to answer questions 11 through 15. This preparation is recommended for the treatment of which condition?

- A) psoriasis
- B) acne
- C) melasma
- D) rosacea
- E) none of the above is correct

Source: hydroquinone is the reference depigmenting agent; salicylic acid improves penetration through exfoliation.

POST-TEST · Question 12 / 15

Vitamin C in the Formulation

Vitamin C is included in this preparation as:

- A)** an antioxidant
- B)** a depigmenting agent
- C)** a penetration enhancer
- D)** all of the above are correct
- E)** none of the above is correct

Source: vitamin C protects hydroquinone from oxidation and improves the stability of the formula.

POST-TEST · Question 13 / 15

Timing of Application

It is recommended to apply this preparation:

- A)** all day and night
- B)** at bedtime
- C)** during the day only
- D)** every 6 hours, day and night
- E)** none of the above is correct

Source: hydroquinone is susceptible to oxidation and photodegradation when exposed to sunlight.

POST-TEST · Question 14 / 15

Hydroquinone Calculation

The amount of hydroquinone required to prepare this formulation is:

A) 3.61 G

B) 4.80 G

C) 5.42 G

D) 6.00 G

E) none of the above is correct

Source: [61 G of fixed ingredients = 90% of the formula; 8% hydroquinone equals 5.42 G]

POST-TEST · Question 15 / 15

Recommended Equipment

It is recommended to prepare this formula using:

- A)** a high-speed mixer
- B)** an analytical balance
- C)** mill
- D)** a good penetration enhancer
- E)** none of the above is correct

Source: hydroquinone is in crystal form; the mill will reduce the particle size.

ANSWER KEY — POST-TEST (1 of 2)

For facilitator use / participant self-assessment · Questions 1–8

Q#	Answer	Rationale
Q1	C	USP <795> defaults non-aqueous liquid/semisolid formulations without supporting stability data to a 90-day BUD; the 180-day default is reserved for solid oral dosage forms such as capsules. Note: this deck's own BUD-table slide simplifies suppositories into the same 180-day row as capsules and topicals — worth revisiting for consistency, but the post-test answer (90 days) matches the confirmed course answer key.
Q2	A	Silicon dioxide is a suspending agent — it distributes the API evenly through the suppository mass. It does not increase solubility, adjust pH, or affect metabolism.
Q3	C	The overage rule adds a mandatory +10% to compensate for losses during molding, scraping, and QC: $30 \text{ suppositories} \times 1.10 = 33$.
Q4	B	Cocoa butter — the most common suppository base — melts at 34–36°C and must be refrigerated (2–8°C); refrigeration is the safe default, even though firm PEG/Witepsol bases can be room-temperature stable.
Q5	E	ACOG and NIH/NICHD recognize compounded vaginal progesterone for all four indications: preventing spontaneous abortion, reducing preterm birth, IVF luteal support, and primary luteal-phase support.
Q6	E	Acne, psoriasis, melasma, and hair loss are all routinely treated with individualized compounded topical preparations.
Q7	D	A 3-roller ointment mill reduces API particle size, and in doing so also produces a smoother, more homogeneous cream with the API evenly distributed through the vehicle.
Q8	B	EMP stands for Electronic Mortar and Pestle — the mechanized device commercially known as the Unguator®.

ANSWER KEY — POST-TEST (2 of 2)

For facilitator use / participant self-assessment · Questions 9–15

Q#	Answer	Rationale
Q9	D	Low-dose naltrexone (1.5–4.5 mg/day) is reported to help Hailey-Hailey disease, refractory psoriasis, and scleroderma, among other immune-modulated conditions.
Q10	D	Minoxidil, finasteride, and dutasteride act through different mechanisms (vasodilation and 5 α -reductase inhibition) and are often compounded together for androgenetic alopecia.
Q11	C	Hydroquinone (a tyrosinase-inhibiting depigmenting agent) combined with salicylic acid for penetration is the classic compounded formula for melasma.
Q12	A	Vitamin C protects hydroquinone from oxidation, preserving the stability and potency of the formula— not a depigmenting or penetration-enhancing role.
Q13	B	Hydroquinone degrades on light exposure (oxidation/photodegradation), so it is applied only at bedtime, paired with strict daytime sun protection.
Q14	C	The fixed-weight ingredients (Vit. C + Betamethasone + Emollient Cream $\approx$ 61 G) make up the other 90% of the batch (100% – 8% HQ – 2% SA): $61 \text{ G} \div 0.90 = 67.8 \text{ G}$ total, then $8\% \times 67.8 \text{ G} = 5.42 \text{ G}$ of hydroquinone.
Q15	C	Hydroquinone is supplied as crystals; a mill is needed to reduce the particle size before it can be evenly incorporated into the cream base.

Each question answered correctly represents demonstration of achievement of the corresponding objective.

References

Bibliography and regulations cited

- USP <795> Pharmaceutical Compounding — Nonsterile Preparations. United States Pharmacopeia. Effective Nov 1, 2023.
- USP <661> Containers — Plastics. United States Pharmacopeia. 2023.
- Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems. 11th ed. Wolters Kluwer; 2014.
- Remington: The Science and Practice of Pharmacy. 22nd ed. Pharmaceutical Press; latest edition.
- Allen, L.V. Jr. Compounding Formulas and International Journal of Pharmaceutical Compounding (IJPC). Latest editions.
- PCCA. Dermatology Compounding Guide. 2023. pccarx.com.
- American Academy of Dermatology (AAD) Clinical Guidelines.
- Act No. 247 of 2004 — Pharmacy Act of Puerto Rico. Puerto Rico Board of Pharmacy.
- Regulation No. 156. Puerto Rico Department of Health. 2022.
- ACPE Standards 2015. Accreditation Council for Pharmacy Education. acpe-accredit.org.

How to Obtain the Certificate

Validation at cfpr.org + CPE Monitor

- 1 Complete the post-test at cfpr.org ($\geq 70\%$ to pass).
- 2 Complete the Activity Evaluation Form.
- 3 Log in at cfpr.org → MY ACCOUNT → COURSE HISTORY.
- 4 Select the course and enter the access code.
- 5 CPE Monitor syncs credits automatically with NABP.
- 6 Save or print the certificate from CPE Monitor.

DEADLINE: (45 days after the event) No exceptions after this deadline.

Thank you for your participation!

Day 2 completed — Basic Non-Sterile Compounding

Contact Information

Lcdo. Jorge L. Reyes Quiñones, RPh

Santa Cruz Compounding Academy

Email: sccapr2025@gmail.com Tel: (787)-632-8319

Colegio de Farmacéuticos de Puerto Rico

PO Box 360206, San Juan PR 00936-0206 · Tel. (787) 754-8649 · cfpr.org

ACPE Provider 0151